



HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)



**CENTRE FOR
AUTOMATION &
ROBOTICS**

DESIGN AND DEVELOPMENT OF A ROBOTIC DISINFECTANT SYSTEM

Concept – Idea



Disinfectant Robot - PATORO



Vapourised Hydrogen Peroxide Robot

OBJECTIVES



Objective 1

To avoid the objects and people around it.
The robot, senses the environment, gets the feedback, and avoids the interference like objects/obstacles this whole system will work on microcontroller and programming of same as per our need.



Objective 2

To spray the disinfecting fluid stored in the tank around the environment with the prefixed actuators in the air with a refreshing smell.





METHODOLOGY

Methodology

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graph TD; Methodology((Methodology)) --> ObstacleAvoidance[Obstacle avoidance:]; Methodology --> SanitizingModule[Sanitizing module:]; ObstacleAvoidance --> U1[Ultrasonic sensor]; ObstacleAvoidance --> A1[Arduino UNO]; ObstacleAvoidance --> MD1[Motor Driver L298N]; ObstacleAvoidance --> DM1[DC Motor]; SanitizingModule --> IS[IR Sensors]; SanitizingModule --> A2[Arduino UNO]; SanitizingModule --> DCP[DC Motor Pump];
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Obstacle avoidance:

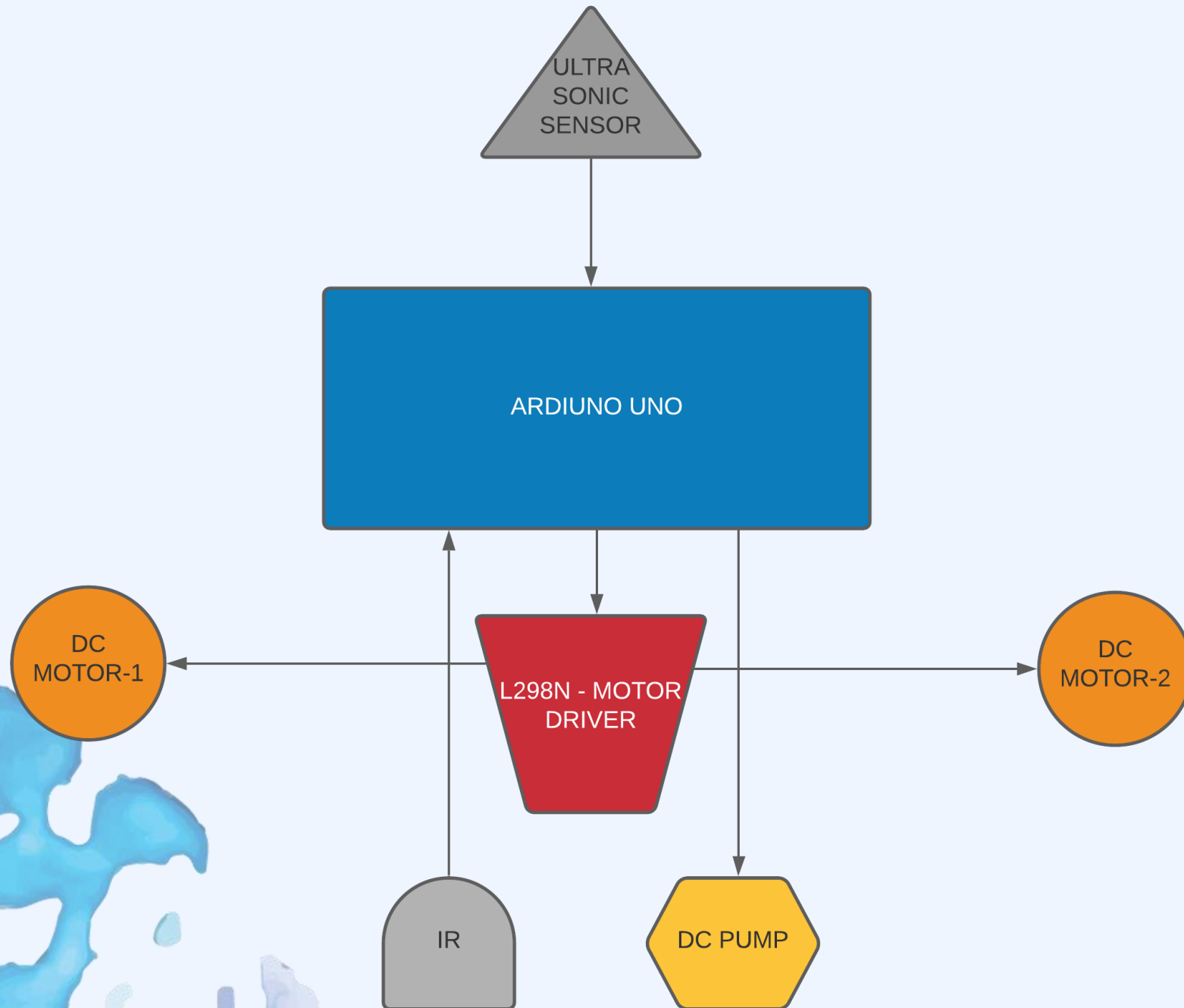
- Ultrasonic sensor
- Arduino UNO
- Motor Driver (L298N)
- DC Motor

Sanitizing module:

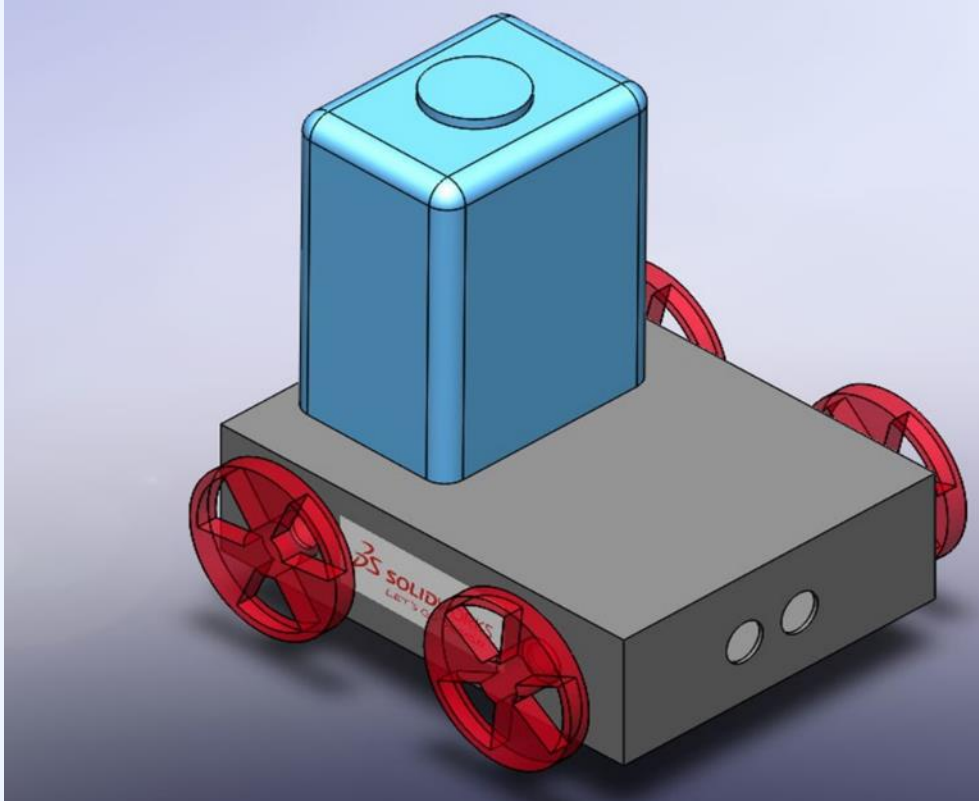
- IR Sensors
- Arduino UNO
- DC Motor Pump



Block diagram

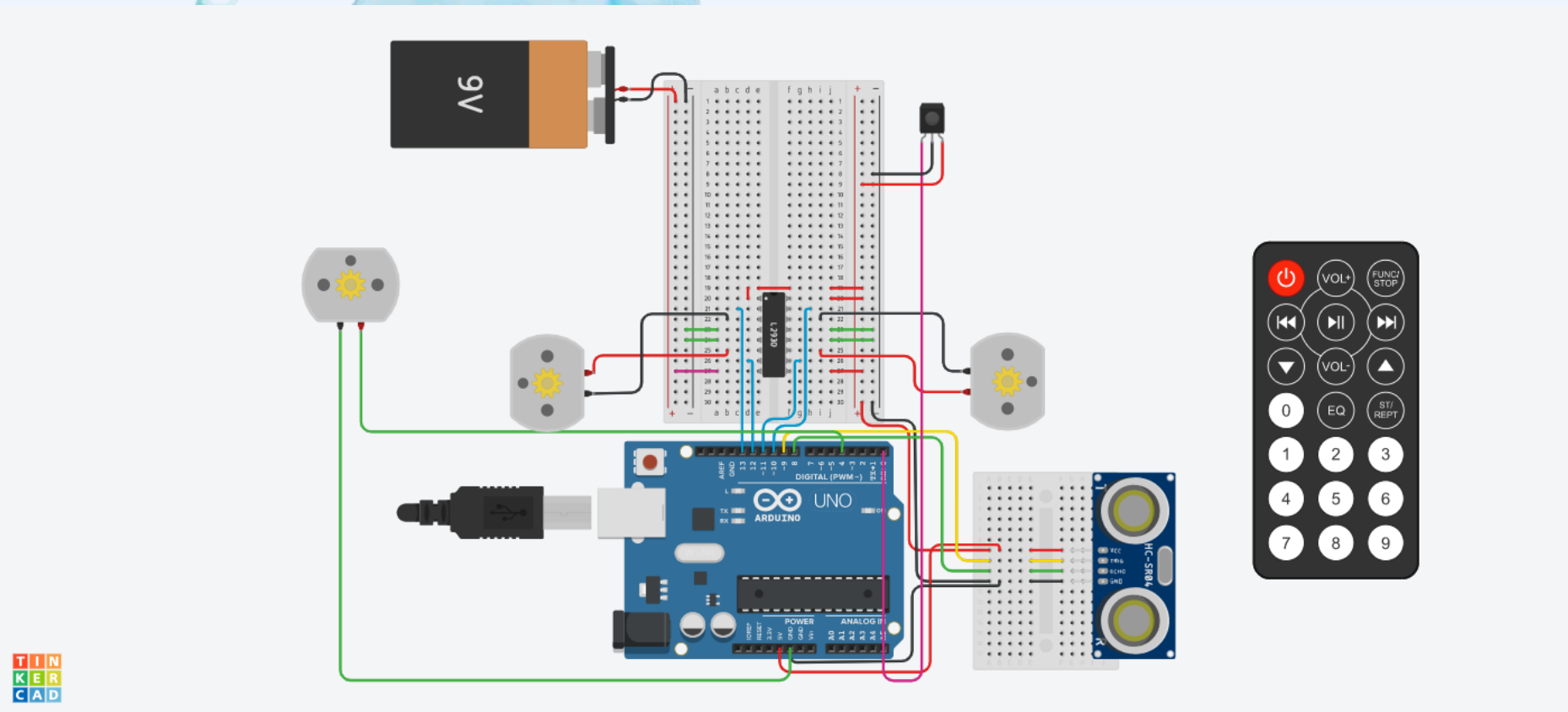


Solid Works



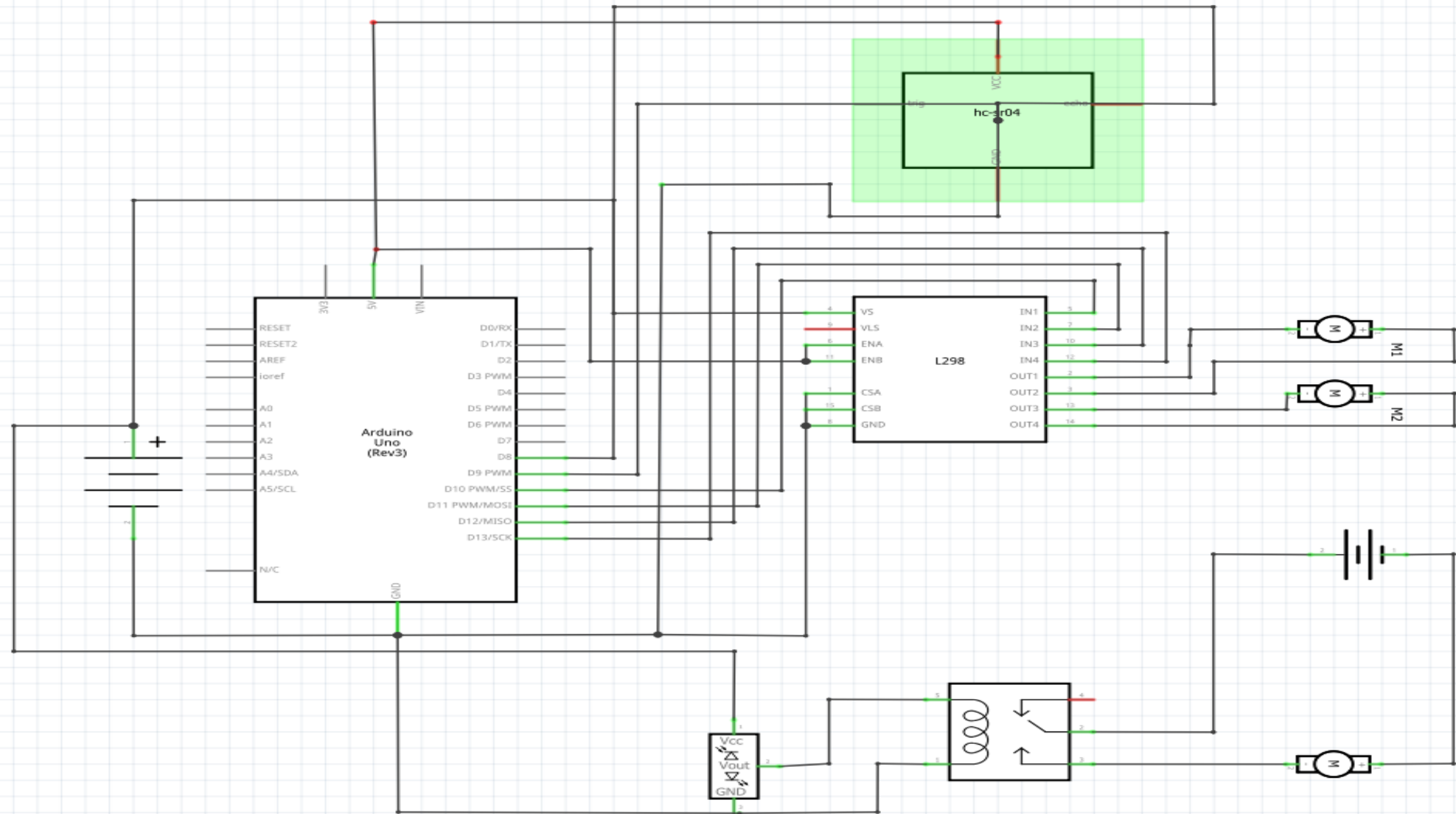
This is the mechanical model created using solidworks which consist of chassis, wheels and sanitizing tank which represents our basic idea how real model looks like with accurate looks and dimension.

Circuit



This is the digital circuit designed in tinkercad representing the functionality of our robotic system which includes obstacle avoidance and sanitizer spraying.

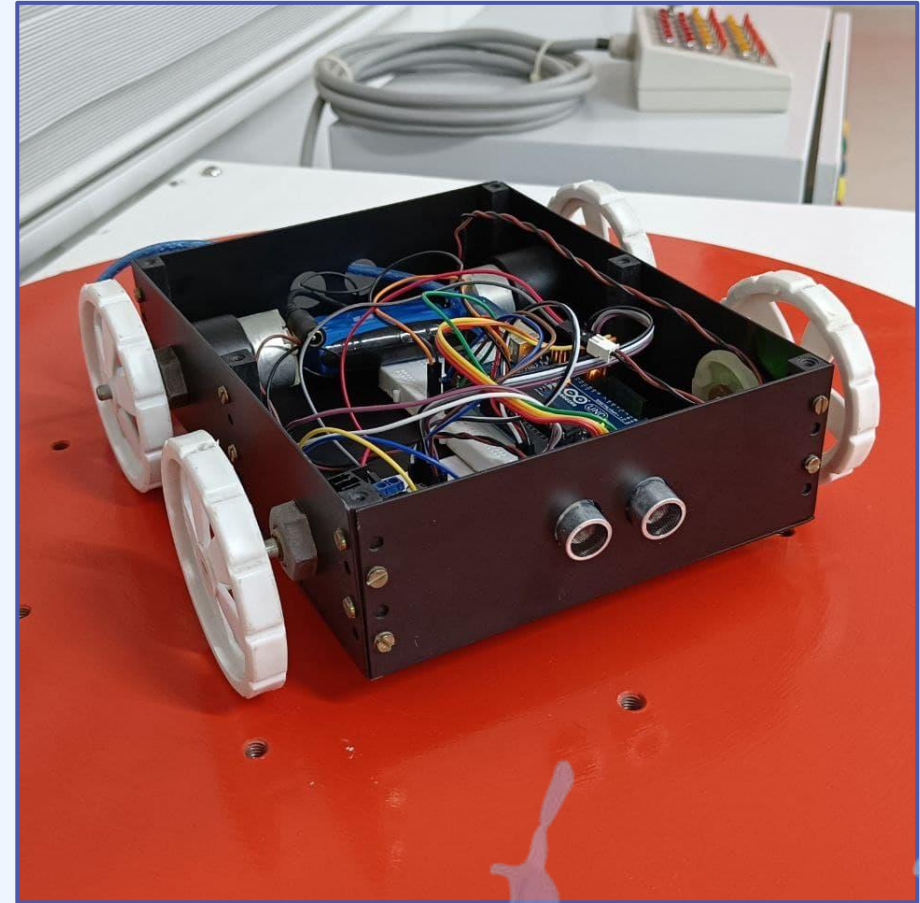
Circuit-block diagram



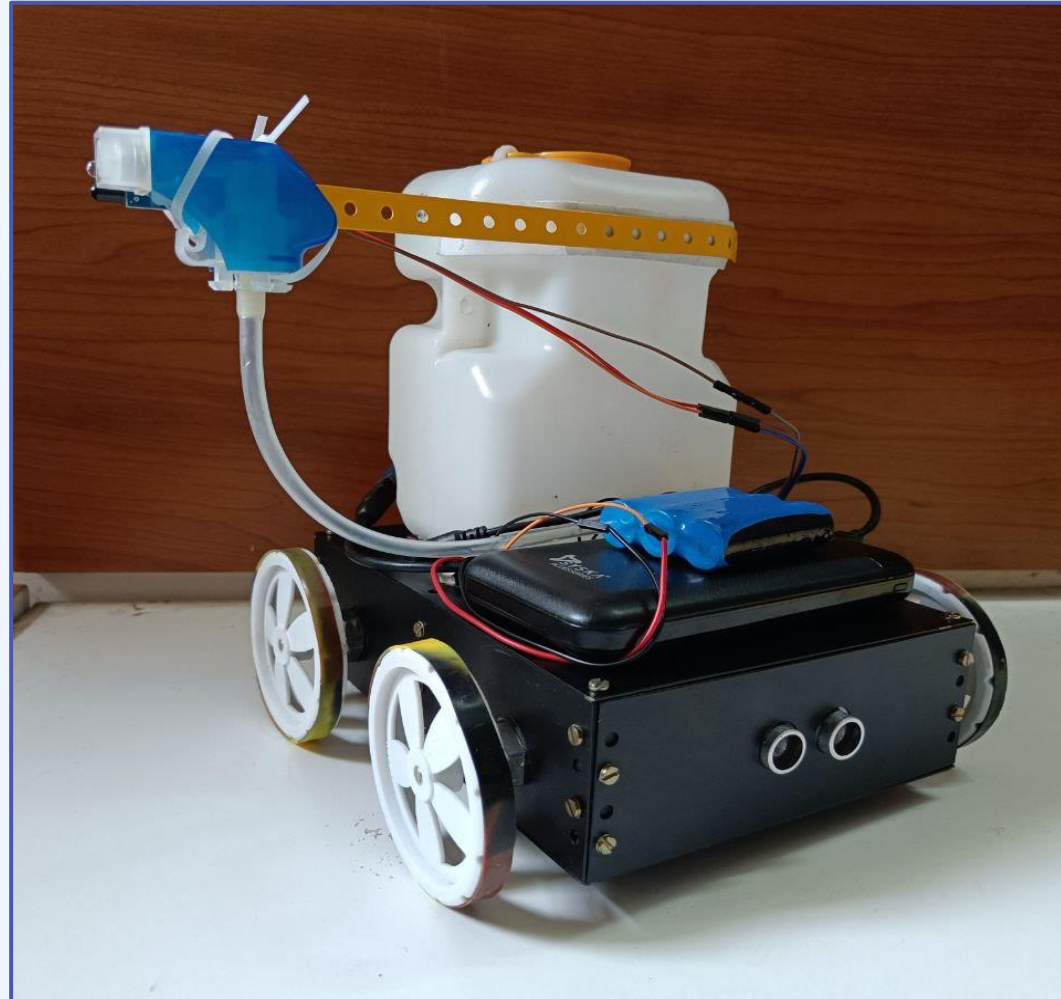
Conclusion

Key points:

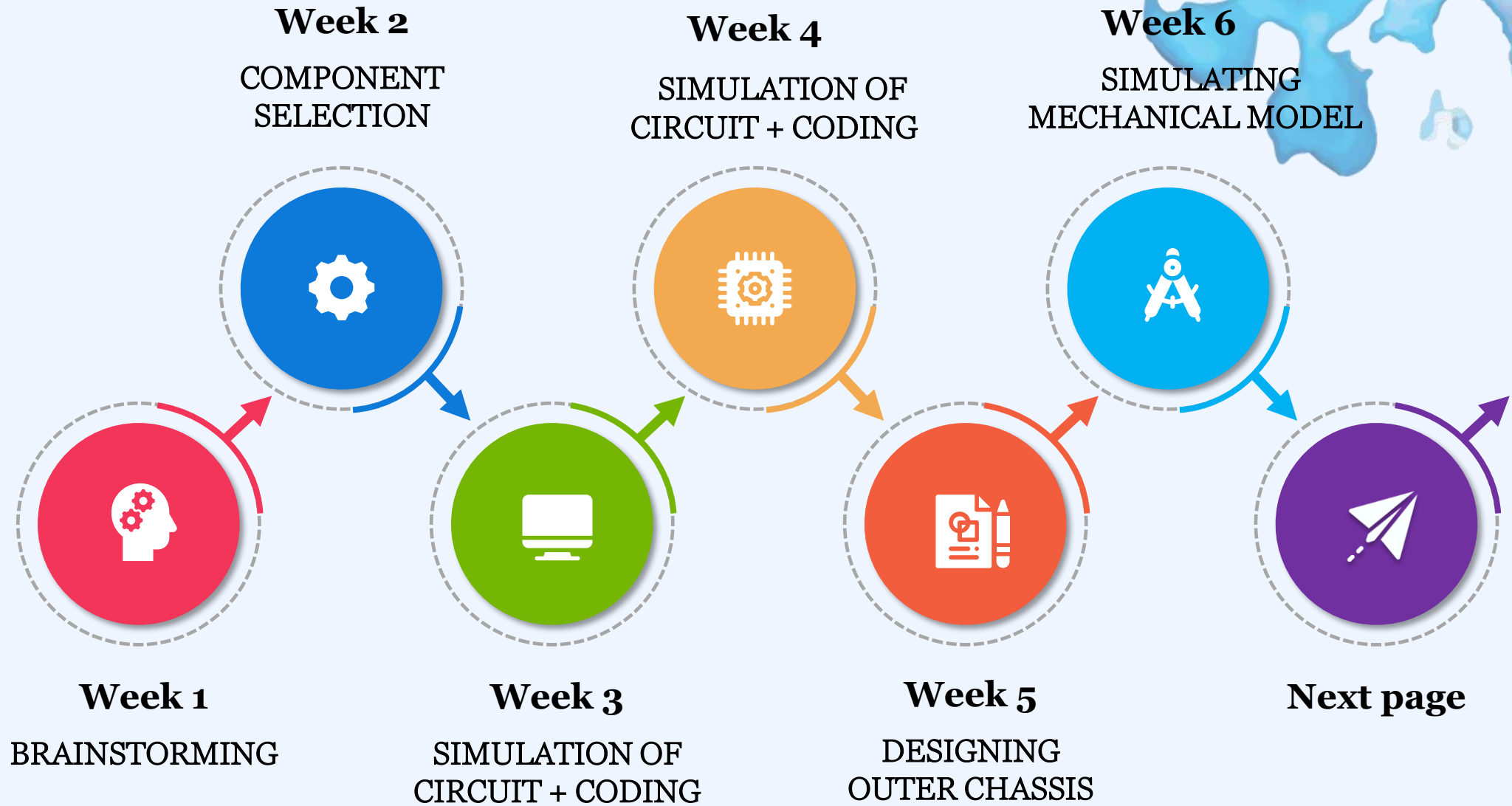
- The robot has been designed and the obstacle avoidance function is successful.
- The simultaneous function of the sanitizing spray is successful.
- The simulation in software and designing in tinkercad and solidworks is completed.
- The designing of chassis and stress analysis is successful.
- The lessons learnt from setbacks are the major learning for future projects.



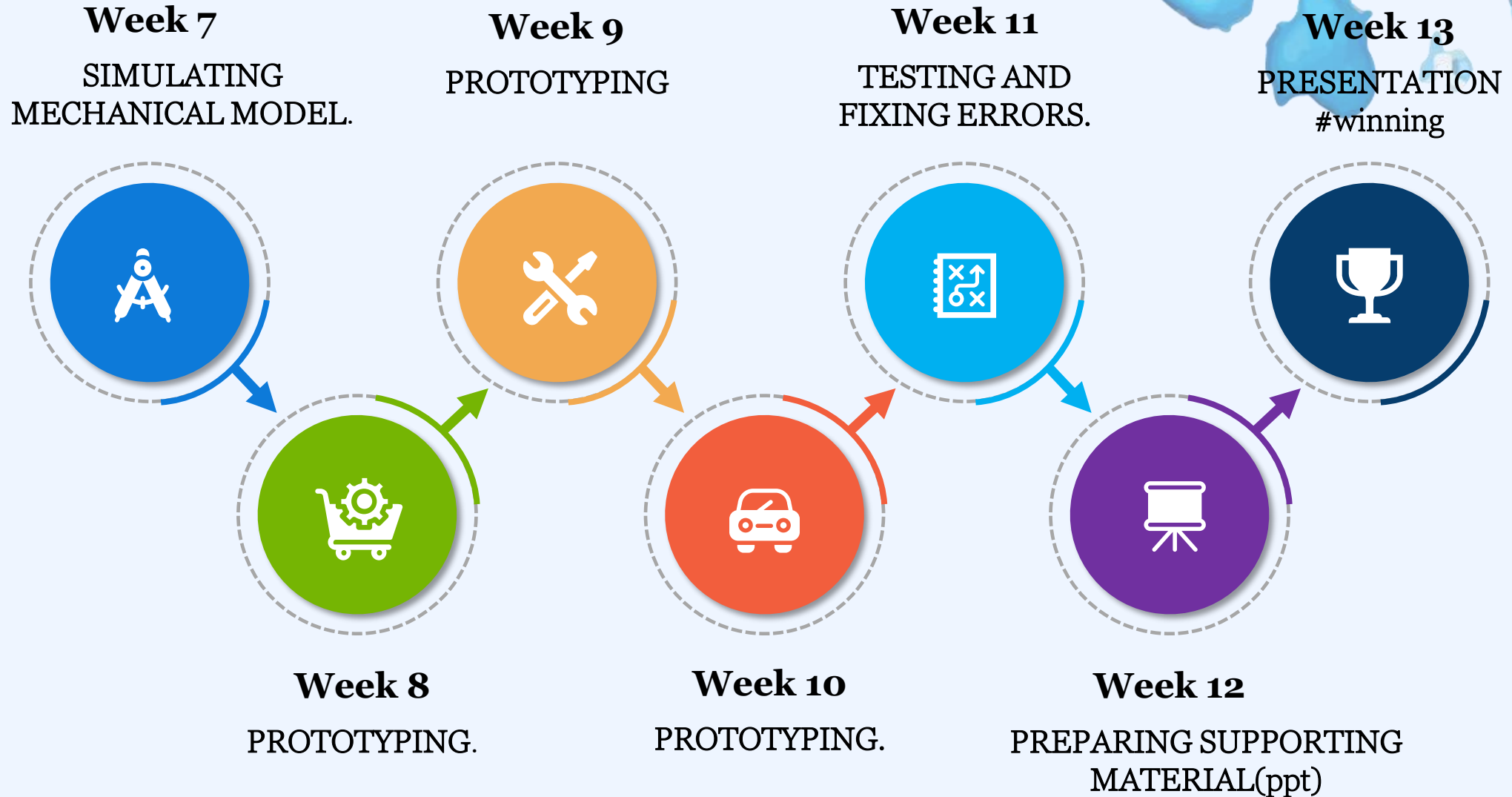
Small scale working prototype



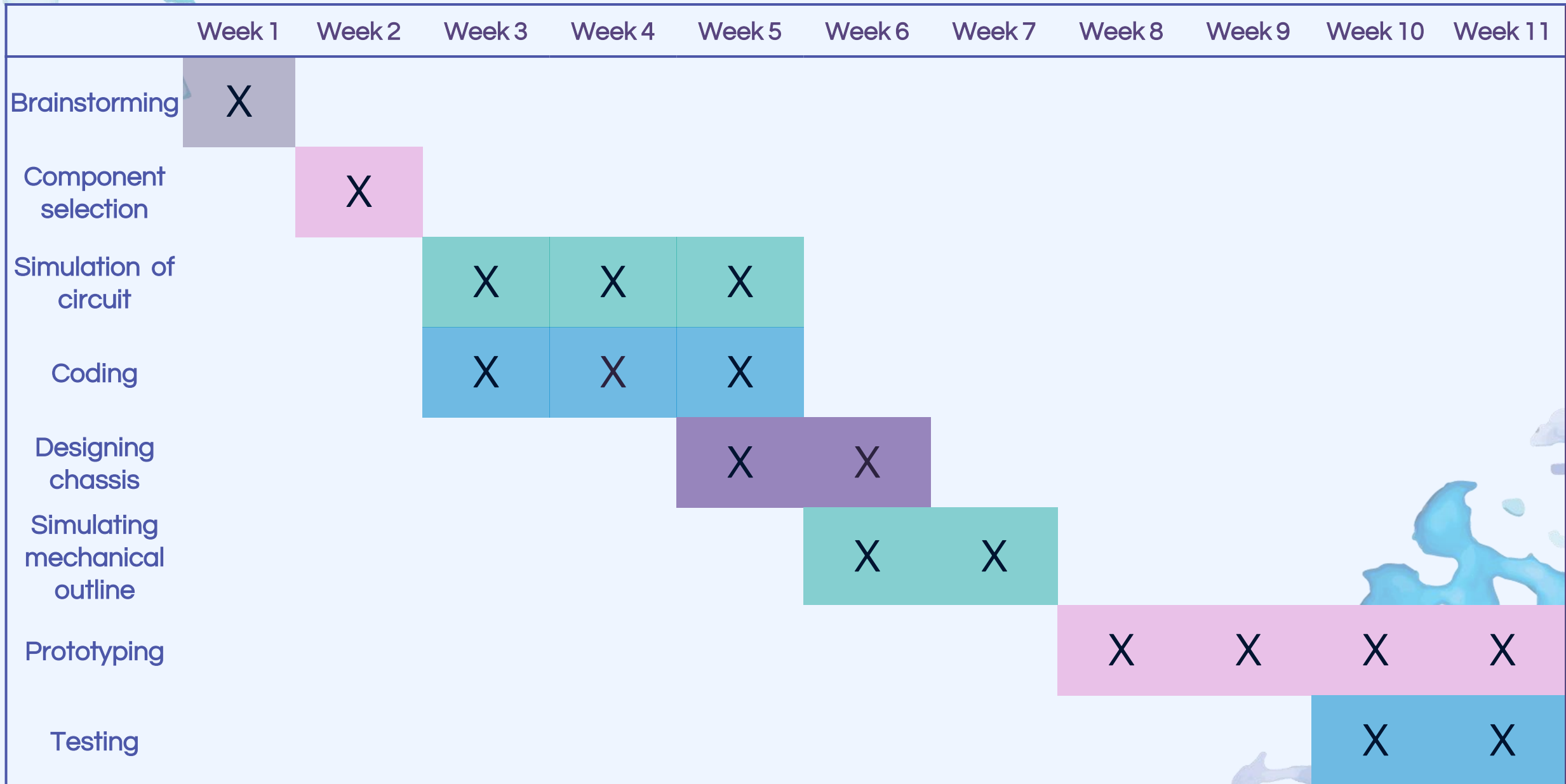
PROJECT PLAN



PROJECT PLAN



GANTT CHART





LITERATURE REVIEW

	JOURNAL	AUTHOR	INFERENCE
	SANITIZATION ROBOT		
01	International Research Journal of Engineering and Technology (IRJET) , Volume: 07 Issue: 08 , Aug. 2020	Apeksha Wadibhasme, Yedhubooshan M M, Kaushik Moolya, Shireen Farhath, Dipti Darade, Sumana Hati	We are learning all the components, their usage and basic idea of creating the system
02	Obstacle Avoidance Robot Using Arduino,International Journal of Engineering Research & Technology (IJERT).Issue - volume6 issue 13, year- 2018.	Pavithra A C and Subramanya Goutham V.	Learning the mechanism of obstacle avoiding design
03	Obstacle Avoiding Robot,Global Journal of Researches in Engineering: H Robotics & Nano- Tech,Volume 17 Issue : 1 Version 1.0 ,Year 2017	Faiza Tabassum, Susmita Lopa, Muhammad Masud Tarek & Dr. Bilkis Jamal Ferdosi.	Learning the mechanism of obstacle avoiding design

	JOURNAL	AUTHOR	INFERENCE
04	Autonomous Navigation and Obstacle Avoidance Vehicle. Written report issued year : 2008	Kyuhyong You	Learning the mechanism of obstacle avoiding design
05	International Journal of Soft Computing and Engineering (IJSCE). Issue :Volume-3, Issue-4, September 2013, pg 52-55	Vivek Hanumante Sahadev Roy Santanu Maity.	Learning distance measurement and working principle.
06	Intelligent tracking obstacle avoidance wheel robot. Kunming university of science and technology,China Issued : 2019	Elsevier B.V	Learning distance measurement and working principle

	JOURNAL	AUTHOR	INFERENCE
07	Design of Disinfection Robot for Livestock Breeding Year of publication: 2013	Qing Chun Feng	We understood the mechanism of spray Nozel and system structure
08	H Bridge DC Motor Driver Design and Implementation with Using dsPIC30f4011 Year of publication: 2017	Tolga Özer , Sinan Kivrak , Yüksel Oğuz	learning the working of h-bridge motor driver
09	Design and Simulation Study of Small Four Wheel Vehicle Chassis for Single Driver Year of publication:2015	M. Z. A Rashid , M. F. Abdul Latif , M. Nur Othman & Marizan Sulaiman	learning the steps to design and test a chasis

PROBLEM STATEMENT

WHO HAS THE PROBLEM?

The most advanced species in the whole universe till date i.e. humans.

WHAT IS THE PROBLEM?

To reduce the spread of virus we need to sanitize regularly touched surfaces.

WHERE IS THE PROBLEM OCCURING?

At high density public places such as train coaches, shopping malls, general stores, etc.

WHY IS IT IMPORTANT TO ADDRESS?

It's important to sanitize the environment at regular intervals to prevent spread of virus.

TEAM 07: ANDRO WIZARDS

DEVASHISH SHARMA (20130032)

SARAN R (20130011)

SAHIL AHAMED SHAIK (20130006)

HEMANG PARAB (20130016)

PROJECT GUIDE:

DR. RM. KUPPAN CHETTY
Associate professor

PROJECT CO-GUIDE:

DR. NAKANDHRAKUMAR
Assistant Professor (SG)



Thanks!

ANY QUESTIONS?